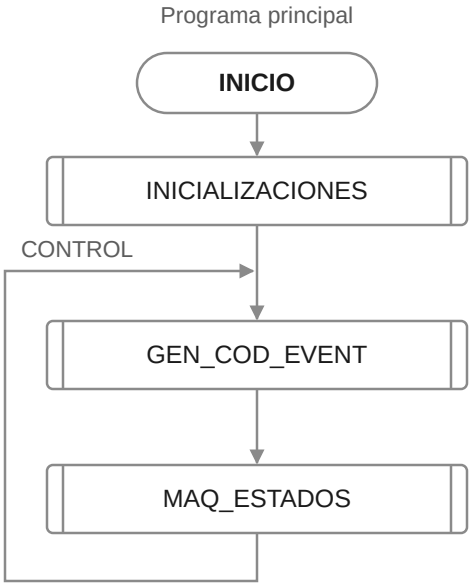
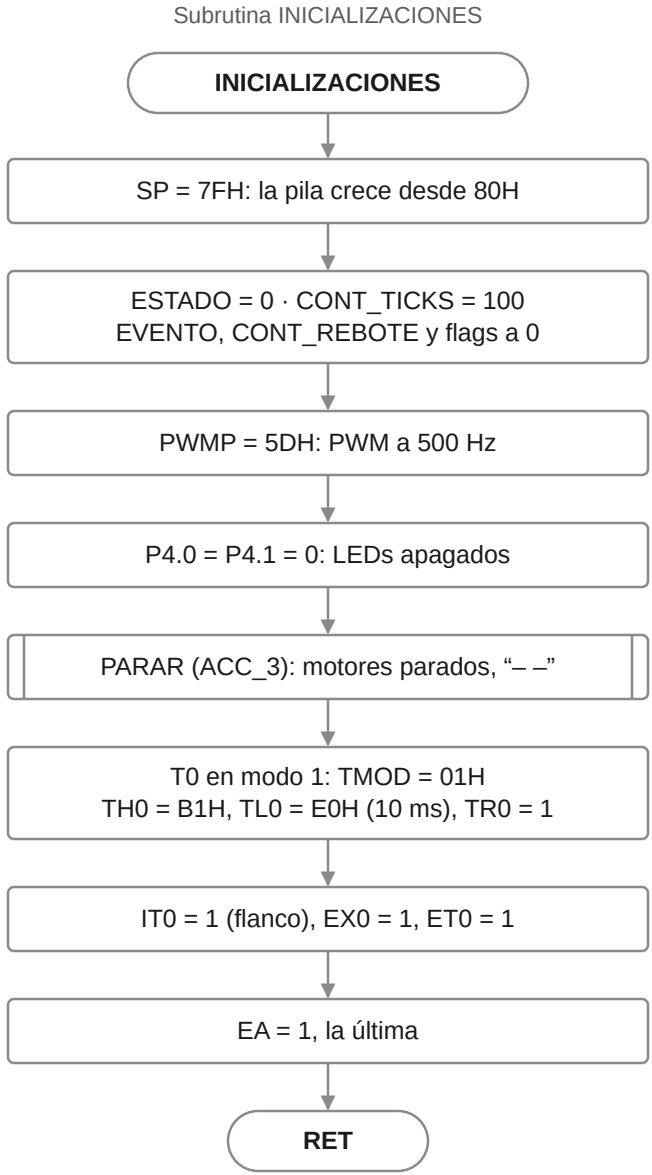
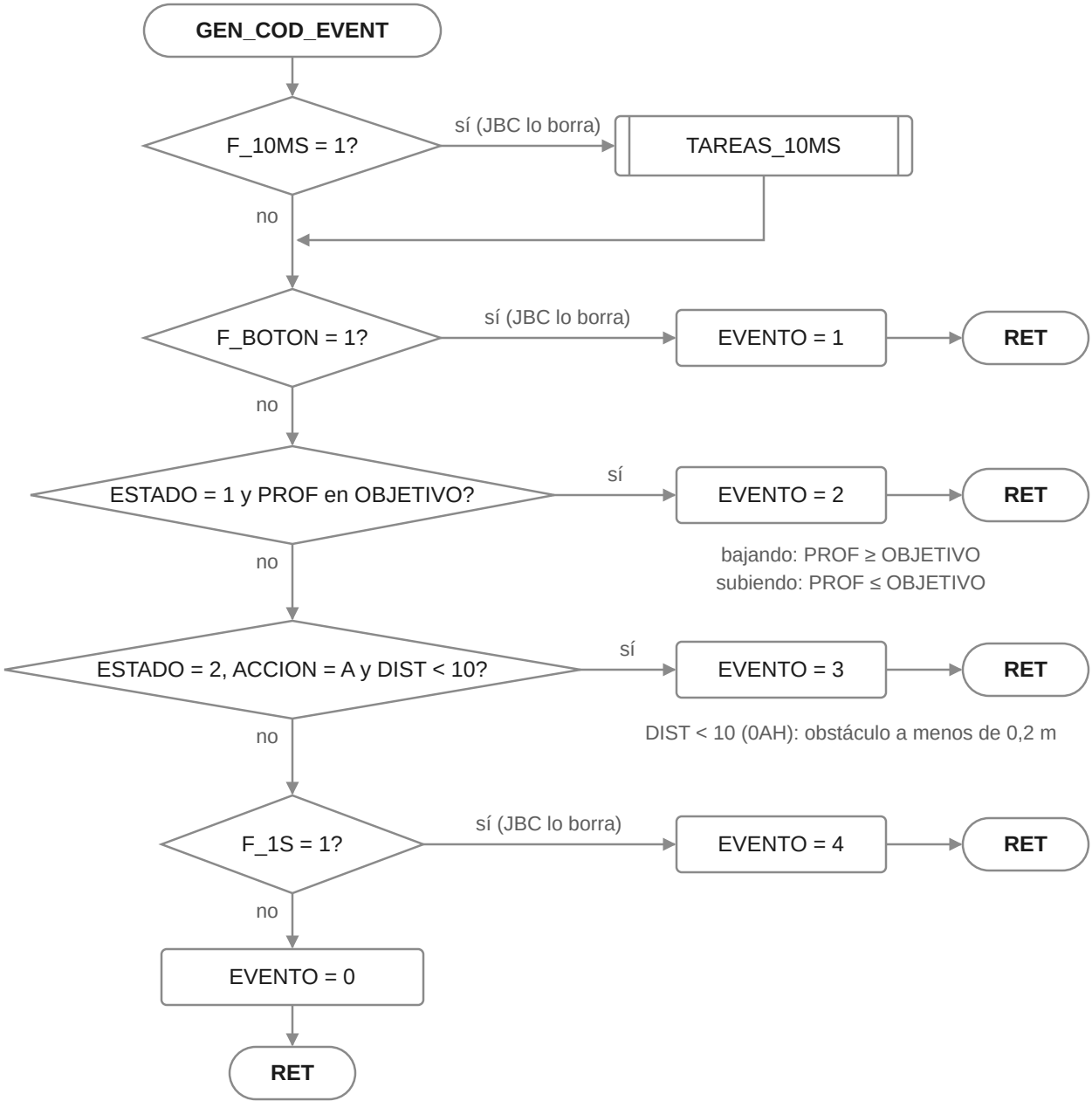
Tras el reset se inicializa una vez y el lazo CONTROL se repite sin fin



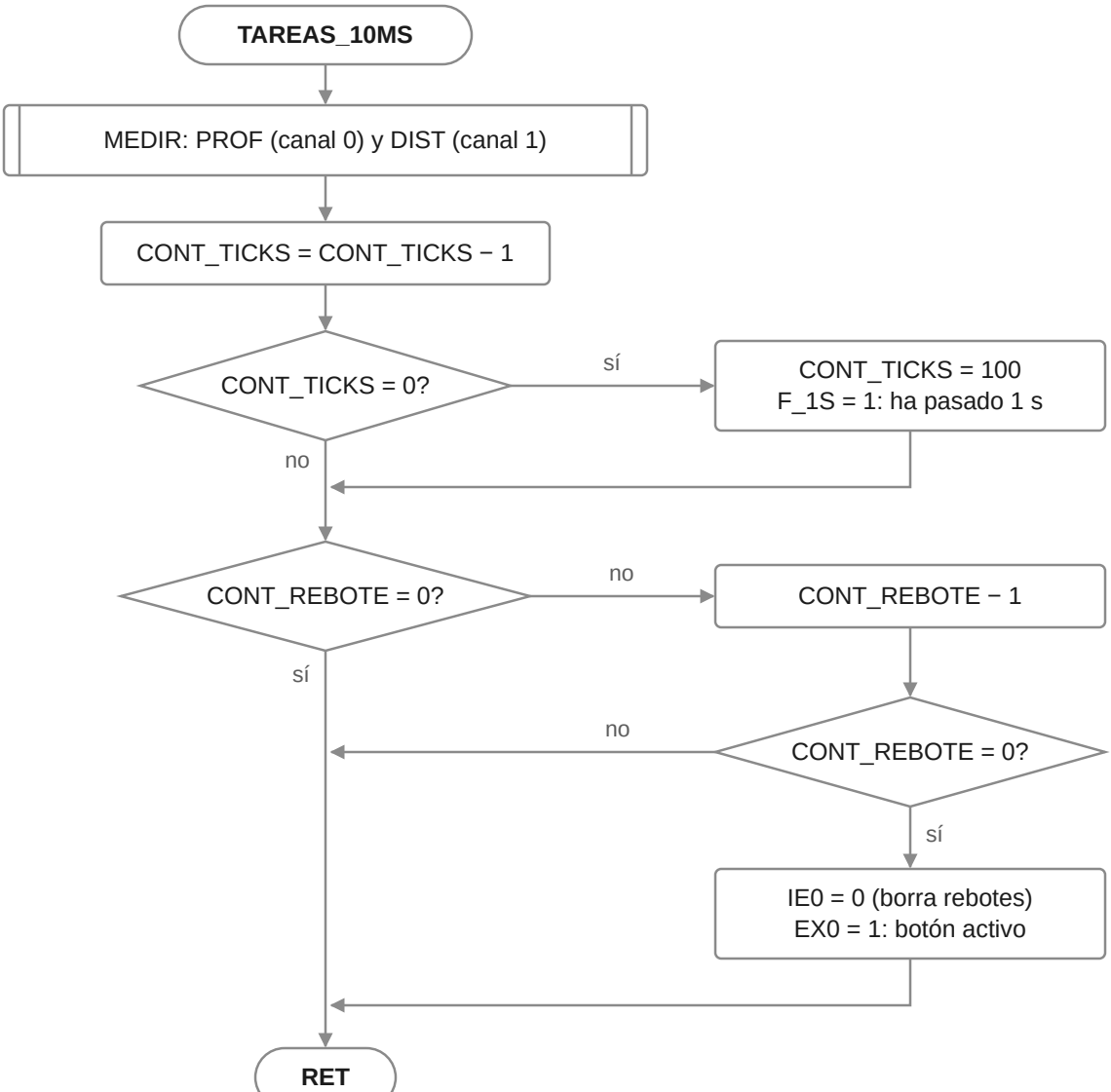
GEN_COD_EVENT deja un código (0–4) en EVENTO y MAQ_ESTADOS lo atiende según ESTADO. El lazo nunca espera.



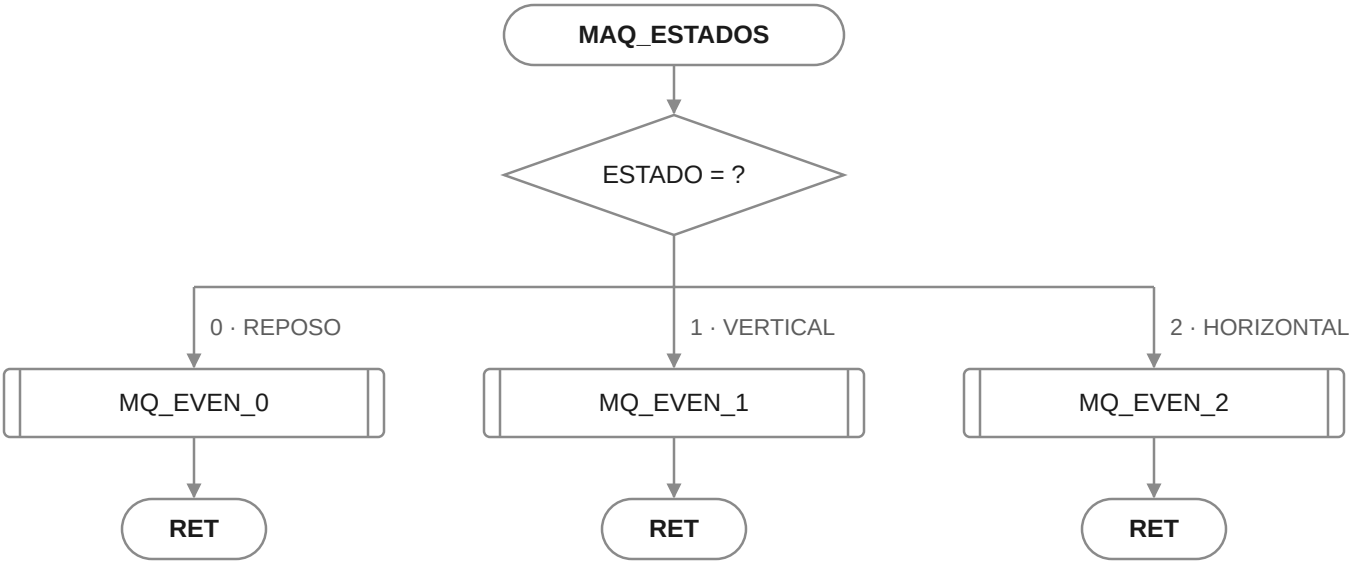
GEN_COD_EVENT deja un solo evento por vuelta: gana el de número más bajo



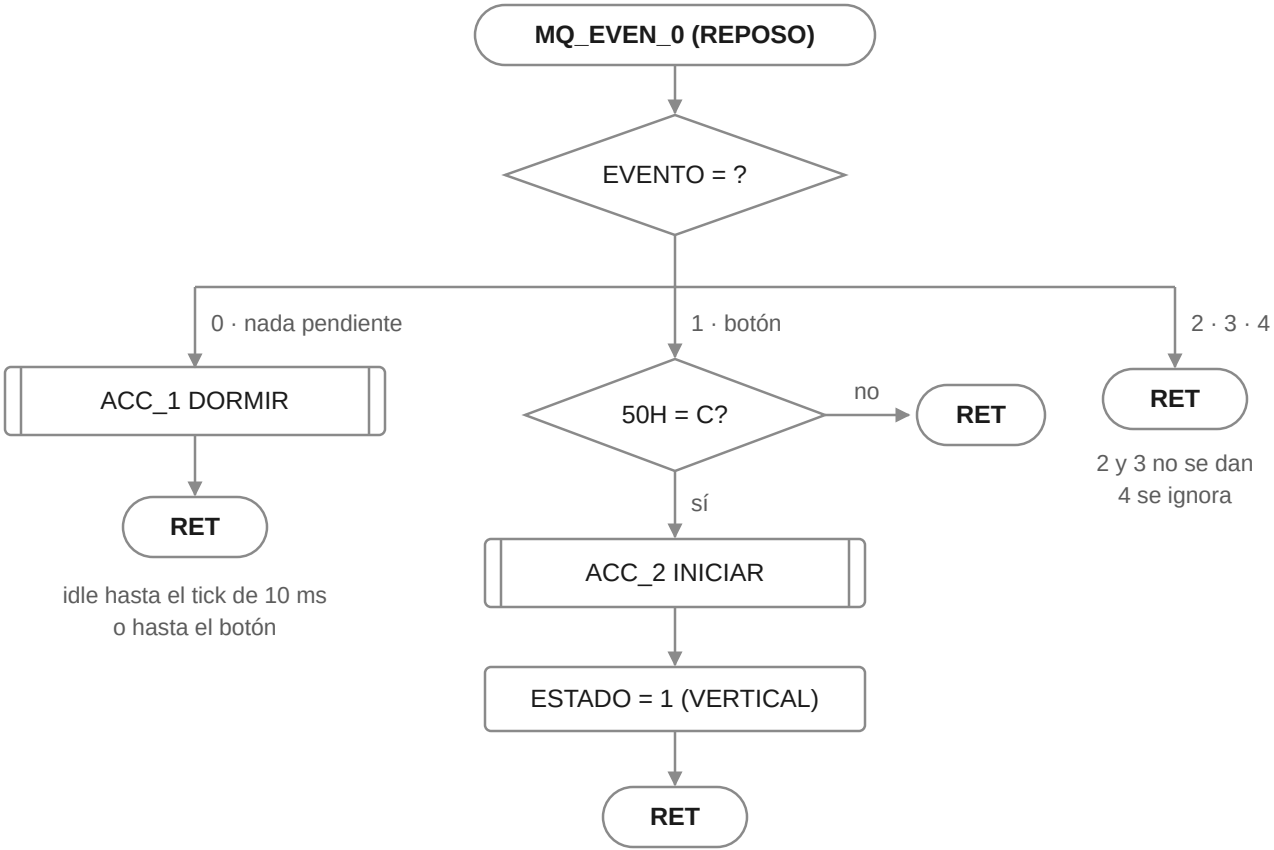
Cada 10 ms: medir, contar el segundo y llevar el antirrebote



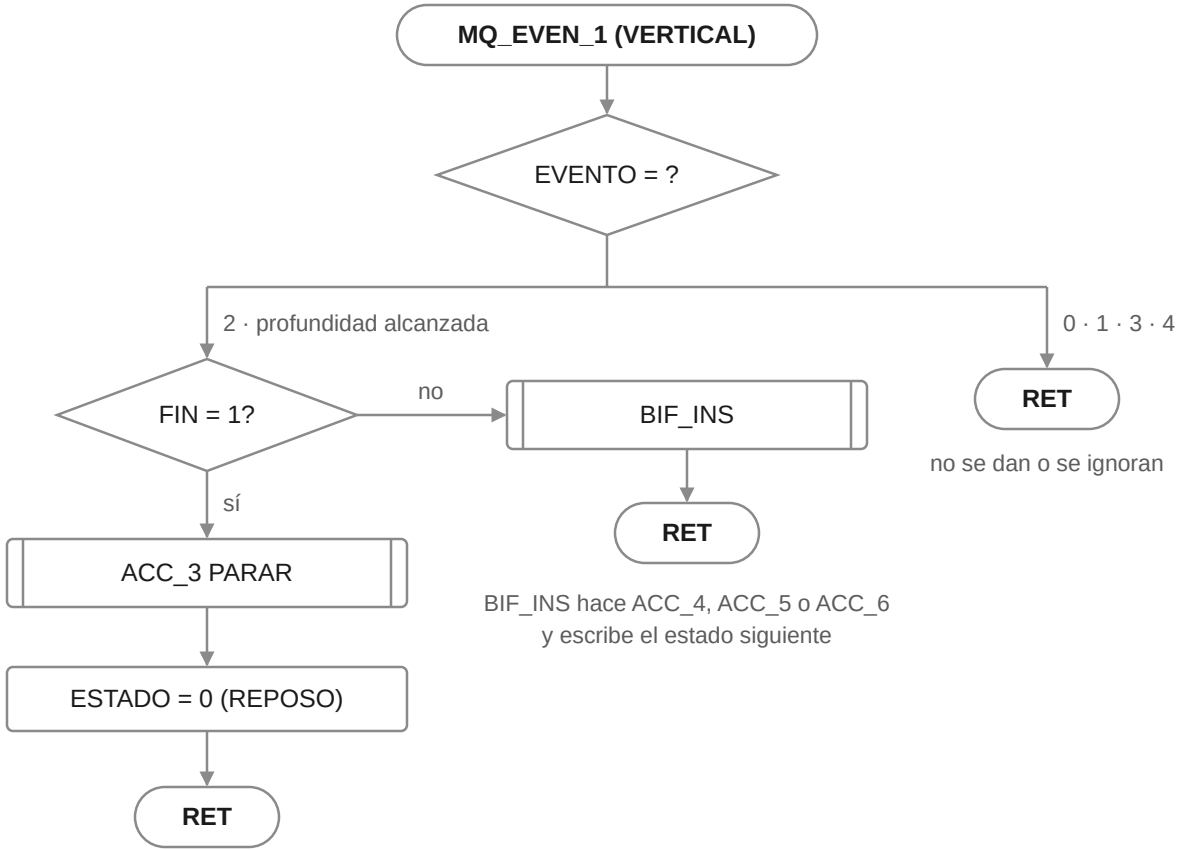
MAQ_ESTADOS salta a la máquina de eventos del estado actual



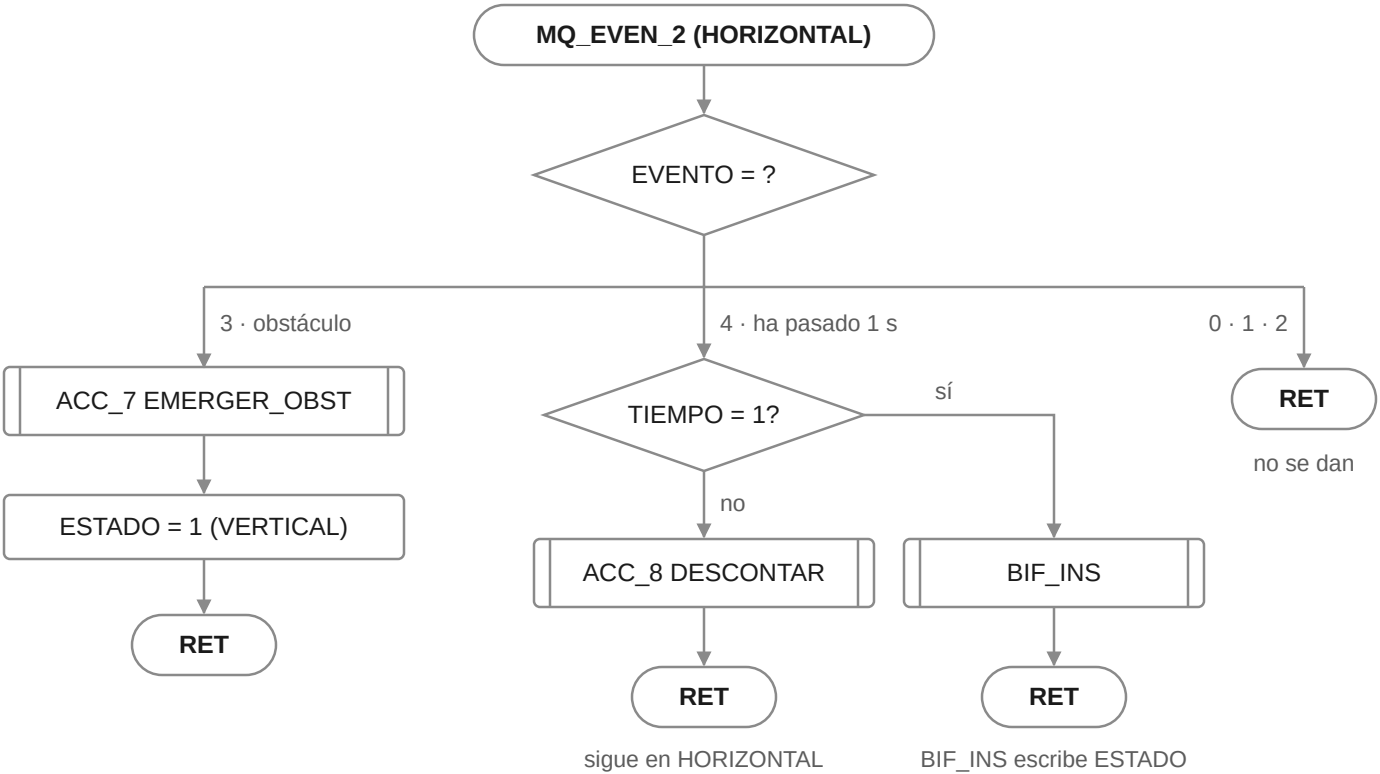
REPOSO duerme si no pasa nada y solo arranca con el botón si 50H es una C



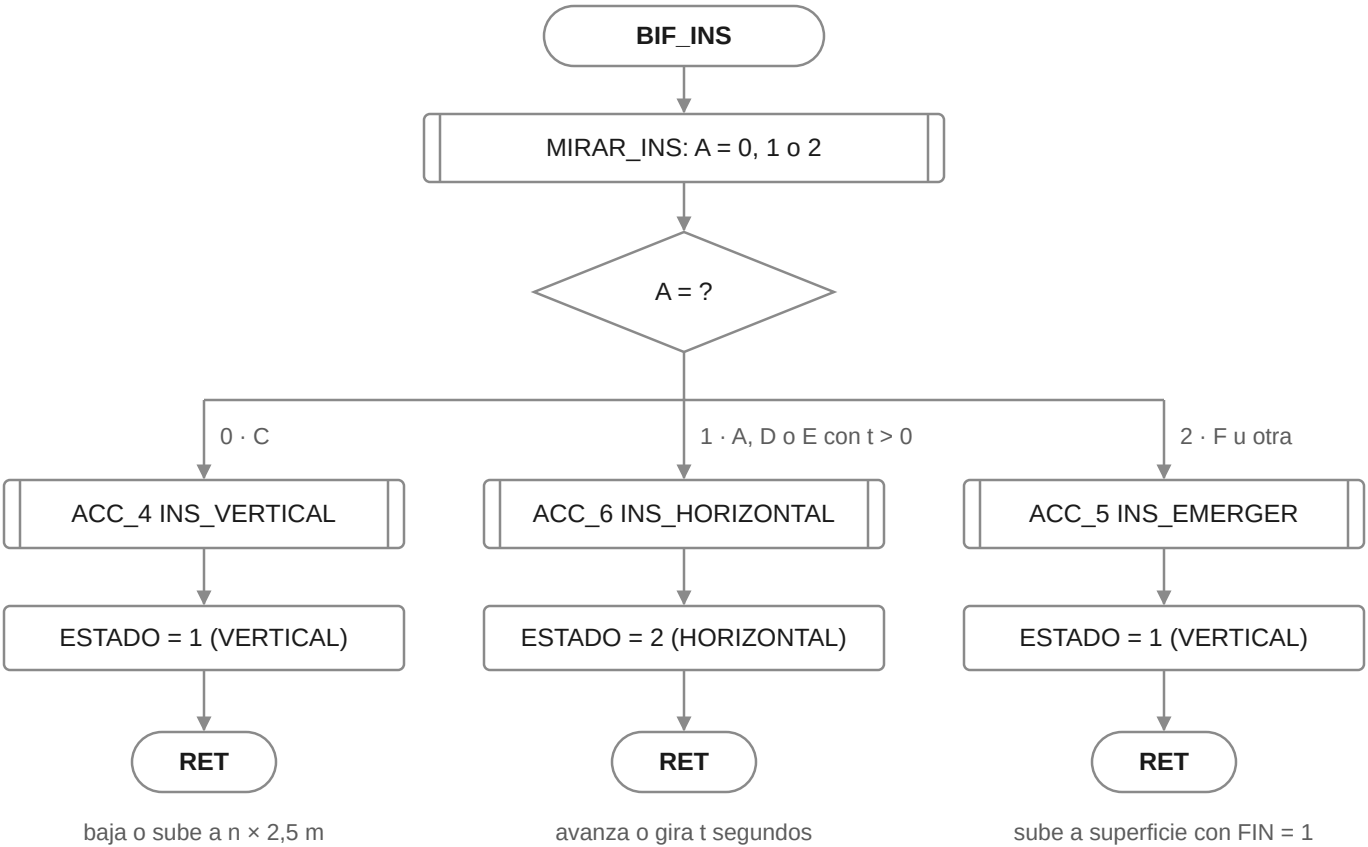
VERTICAL solo atiende el evento 2: profundidad alcanzada



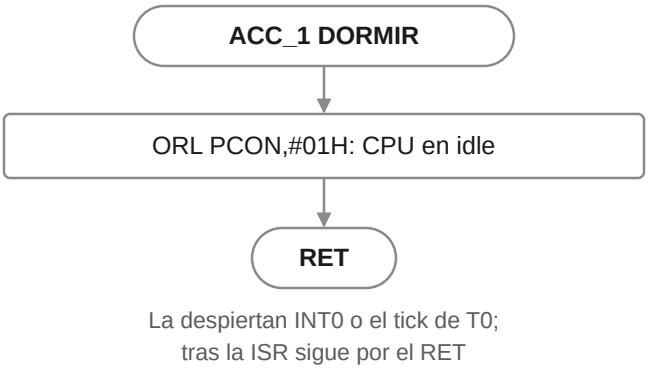
HORIZONTAL atiende el obstáculo (3) y el paso de cada segundo (4)

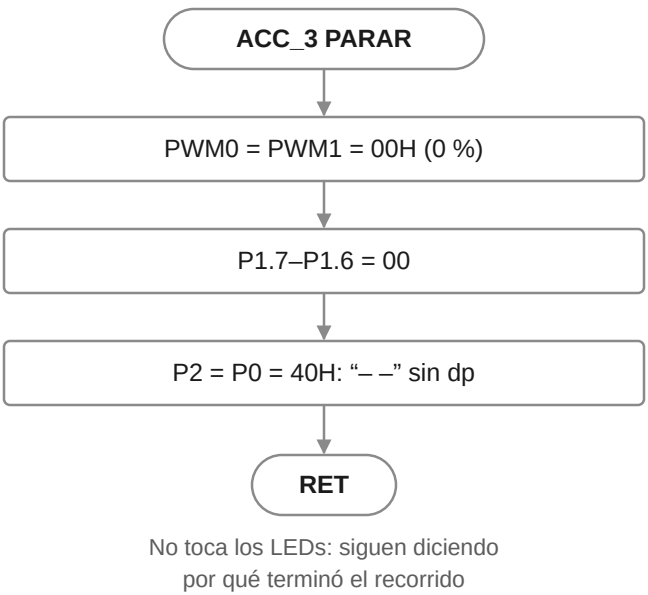


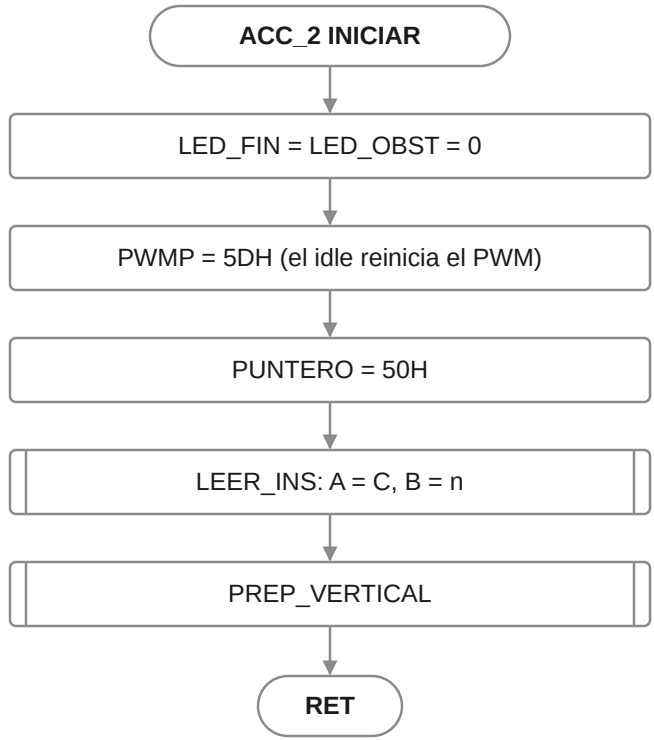
BIF_INS es el ● del diagrama de estados: la siguiente instrucción decide el estado

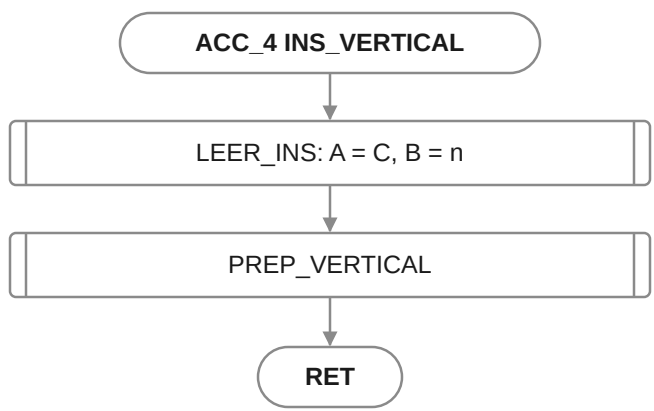


ACC_1 a ACC_4: cada acción fija sus salidas y vuelve sin esperar

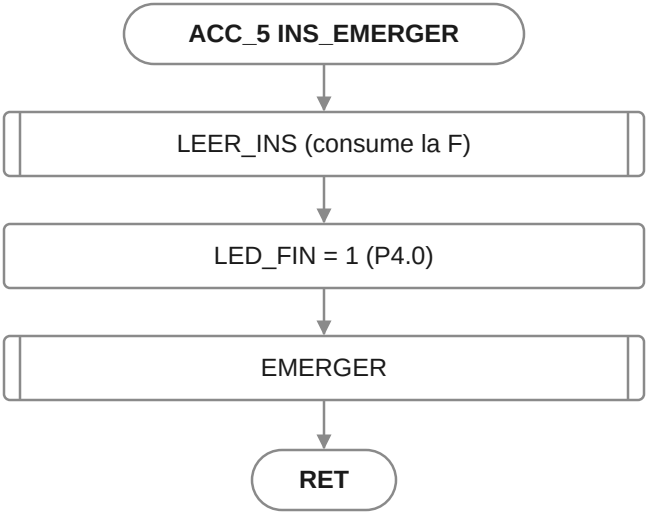


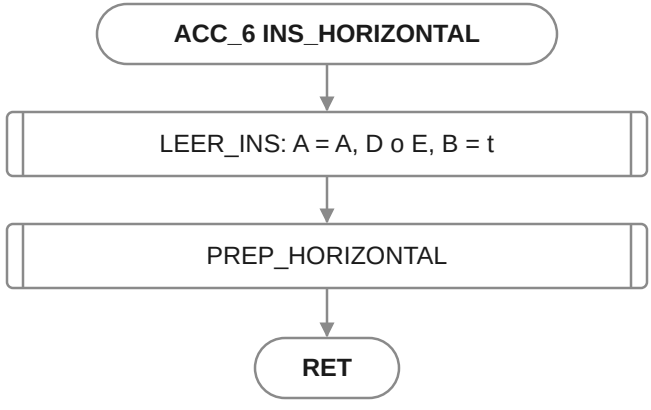


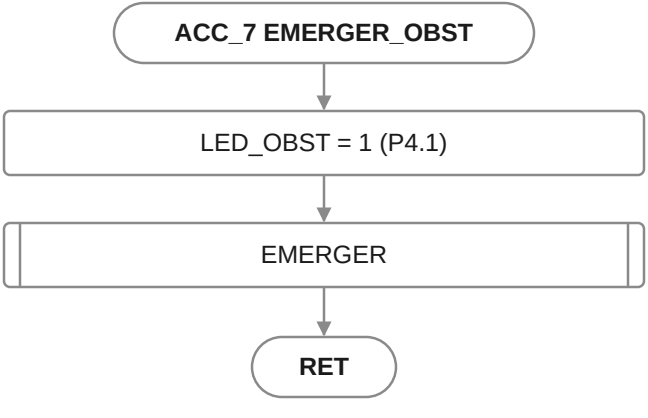


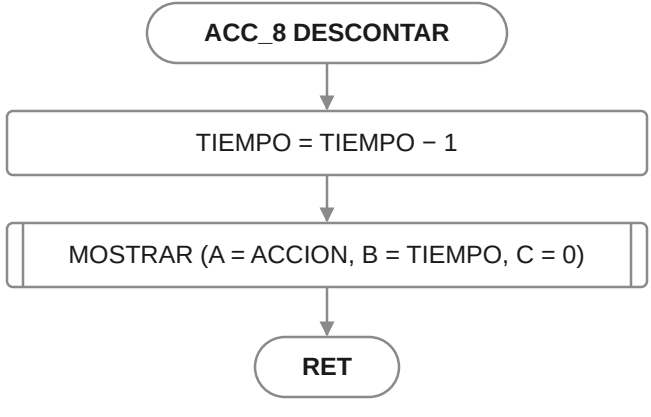


ACC_5 a ACC_8: subir, avanzar o girar, abortar y descontar



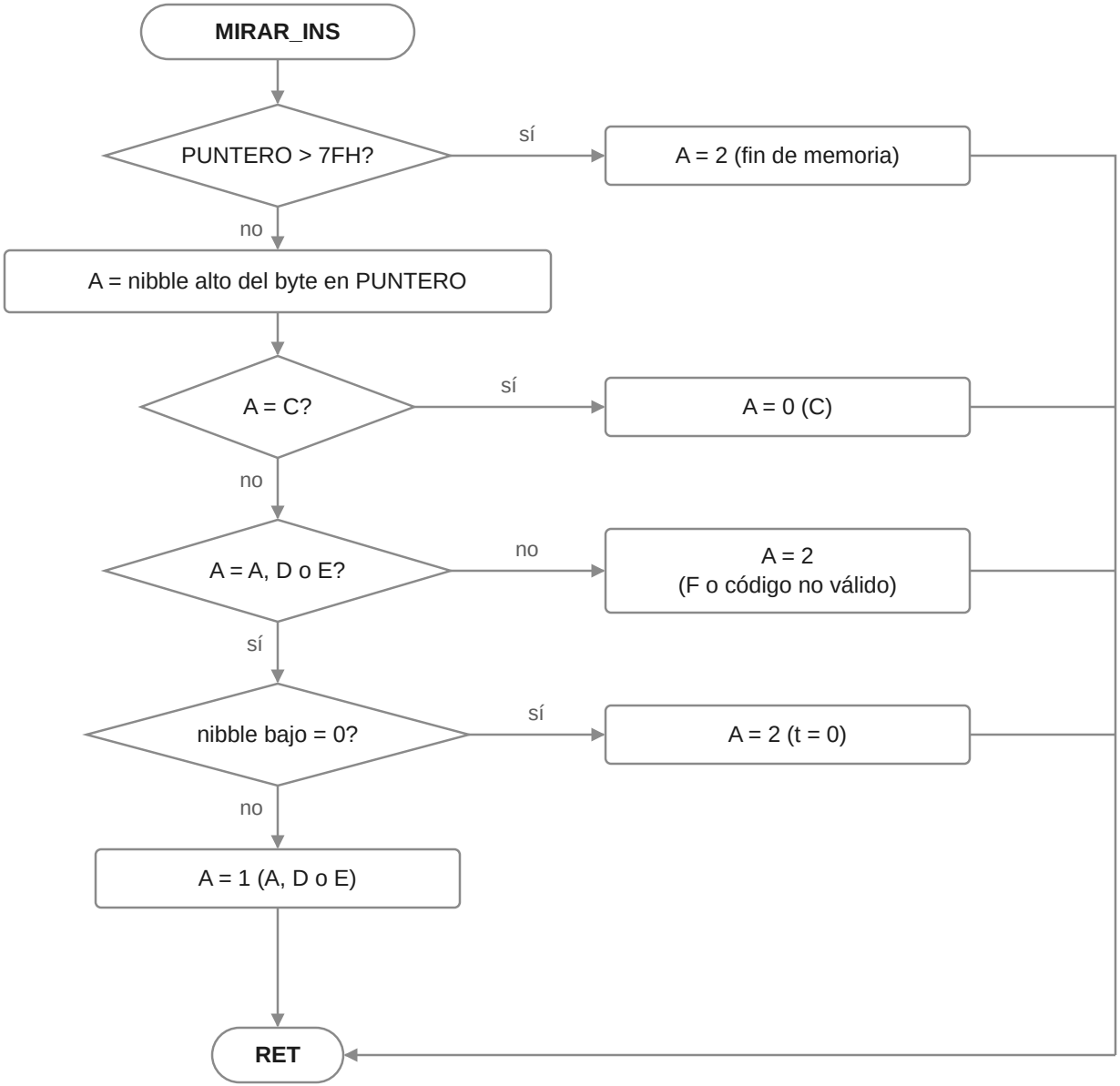




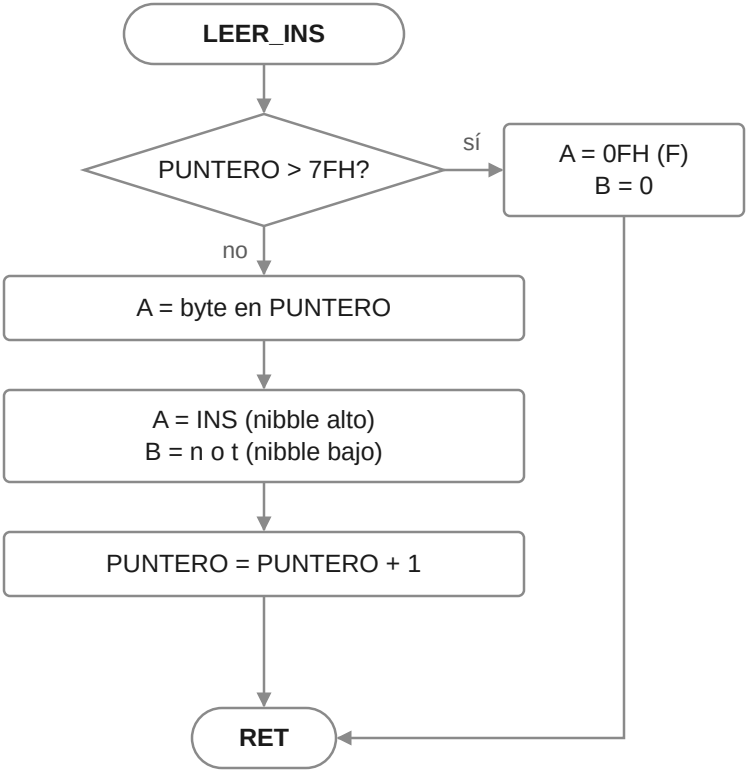


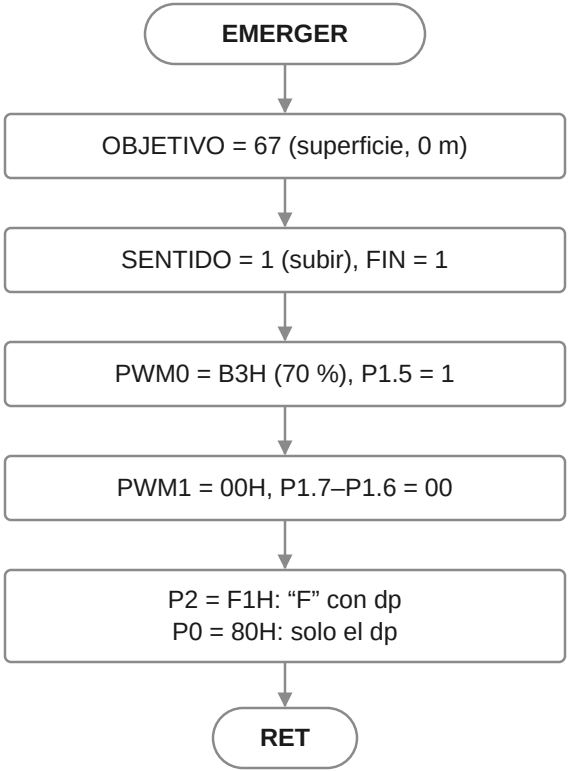
refresca la letra y los segundos, sin dp

MIRAR_INS clasifica la siguiente instrucción sin mover PUNTERO

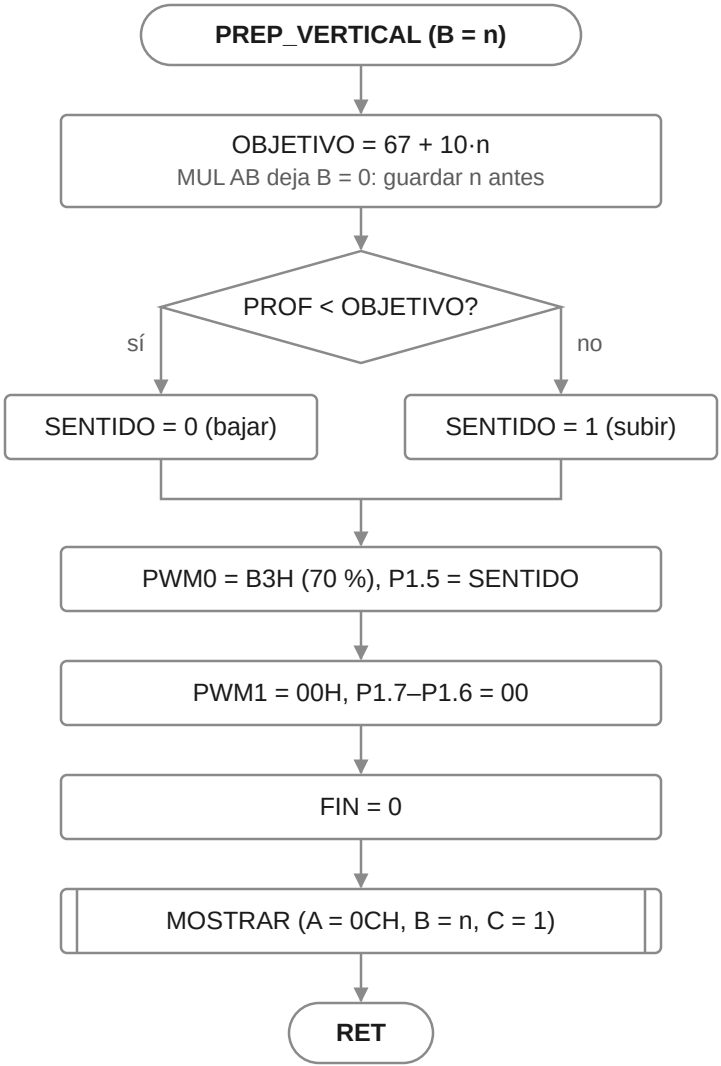


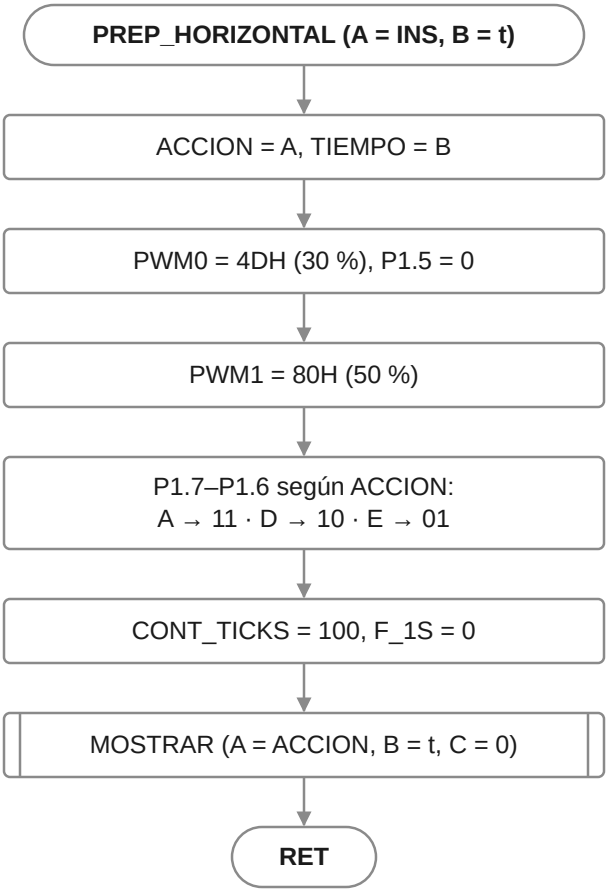
LEER_INS consume la instrucción y EMERGER prepara la subida a 0 m





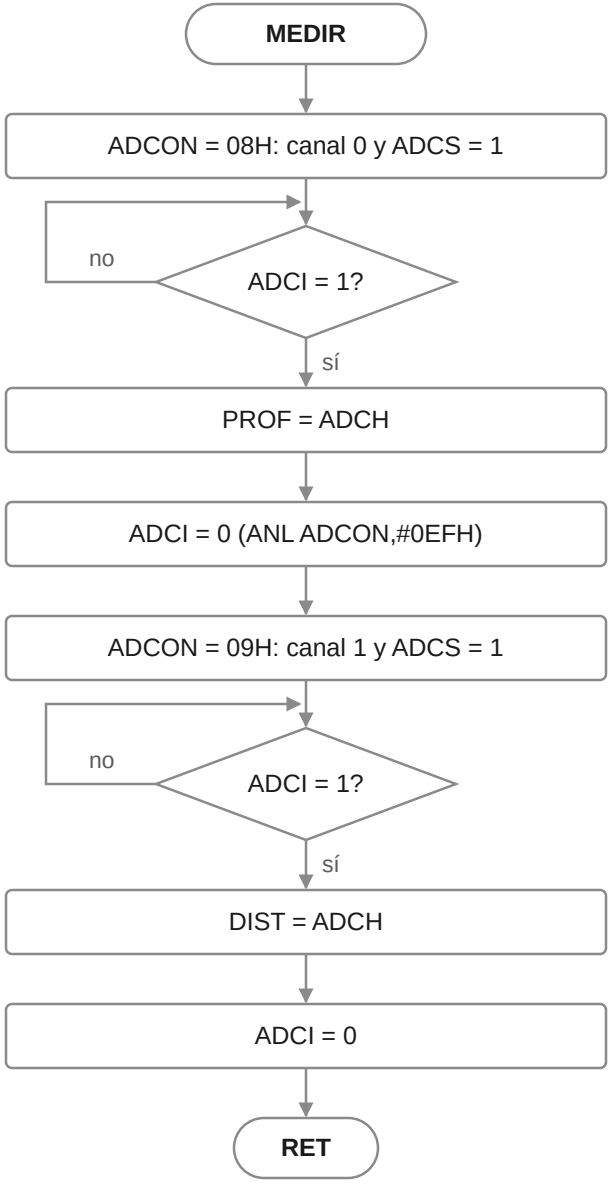
PREP_VERTICAL y PREP_HORIZONTAL escriben todas las salidas de la maniobra



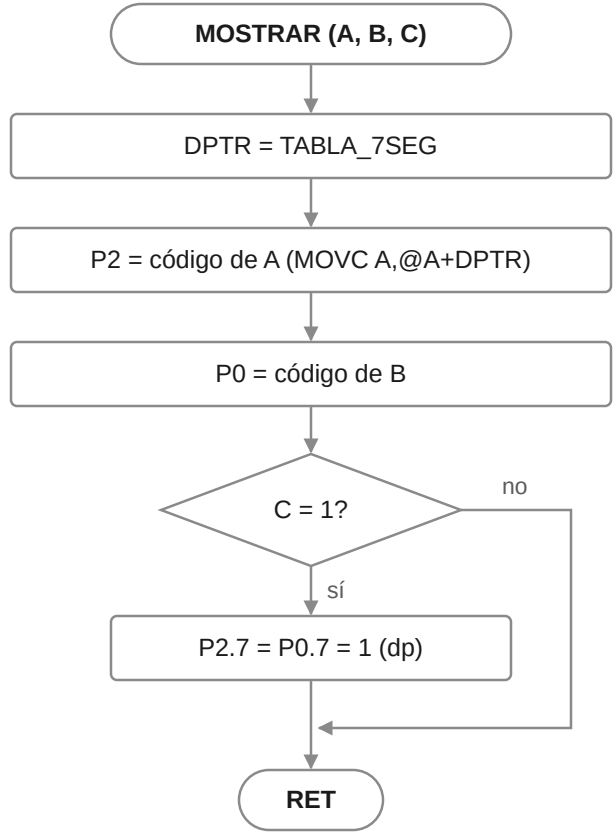


CONT_TICKS = 100: el primer segundo es completo

MEDIR convierte los dos canales seguidos y MOSTRAR traduce con TABLA_7SEG

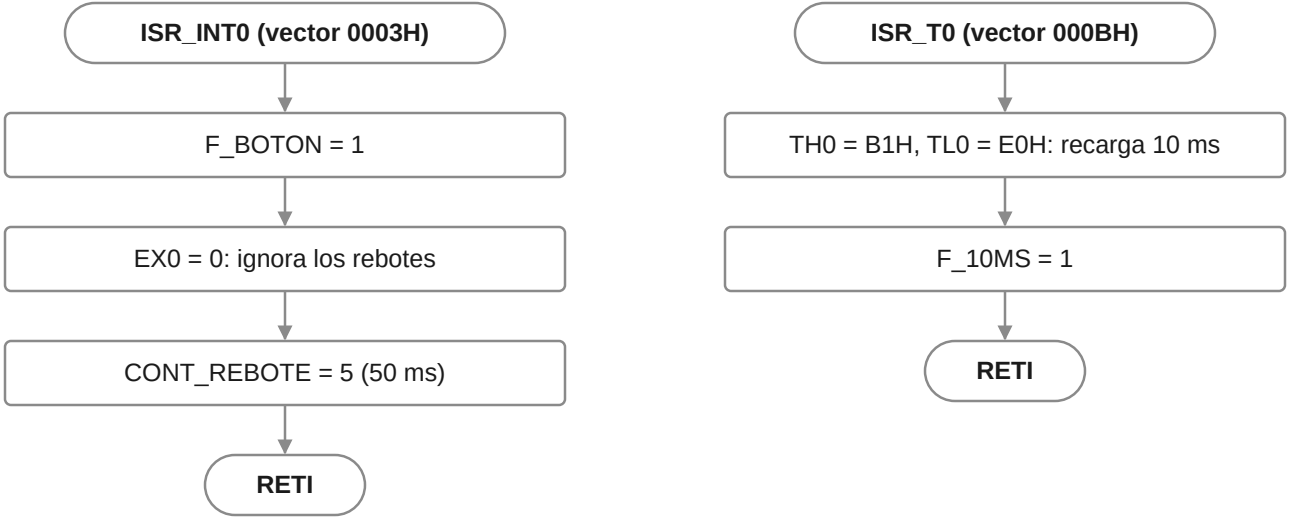


cada espera dura unos 25 µs (50 ciclos máquina)



A = letra: 0AH, 0CH, 0DH, 0EH o 0FH
B = número de 0 a F · C = 1 si lleva dp
TABLA_7SEG: 16 códigos, de 0 a F

Las dos interrupciones solo levantan flags y vuelven con RETI



IE0 y TF0 los borra el hardware al saltar al vector.
Ninguna toca A ni PSW, así que no hacen falta PUSH ni POP.